

Dr. Saif Ali Khan

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Education

Ph.D. in Statistics

Jul 2022 - Nov 2025

Babasaheb Bhimrao Ambedkar University, Lucknow

Title: *Infectious Disease Modeling through Distinct Statistical & Machine Learning Techniques*

M.Sc. in Statistics

2018–2020

Aligarh Muslim University, Aligarh

GPA: 7.690/10.00

Dissertation: Image Classification of Skin Lesion

B.Sc. (Hons) in Statistics

2015–2018

Aligarh Muslim University, Aligarh

GPA: 7.843/10.00

Dissertation: Effectiveness of Choice-Based Credit System (Survey Report)

Professional Qualifications

• GATE (Data Science & AI) (2024)

• O Level (NIELIT) (2022)

Experience

Assistant Professor

Nov 2025 - **present**

Directorate of Online Education

Manipal Academy of Higher Education, Manipal

Research Assistant (UPCST funded project)

Jun 2023 - Oct 2024

Project title: Statistical modeling for COVID-19 spread using different approaches

Department of Statistics, BBAU, Lucknow.

- Designed compartmental and statistical models for COVID-19 transmission
- Applied time series and ML models for prediction

Research Interests and Skills

- Data Science & Predictive Analytics, Machine Learning, Time Series Analysis, Statistical Modeling, Mathematical Epidemiology
- R/Python, Excel, Jamovi

Publications

1. Khan, M. M. A., Batool, T., Ali, A., **Khan, S. A.**, Naeem, M., & Mukarram, M. (2026). Microwave-induced eustress in *Datura innoxia*: Integrated photosynthetic, oxidative, and cellular responses to a low-dose hormetic elicitor. *Biocatalysis and Agricultural Biotechnology*, 76, 104207. <https://doi.org/10.1016/j.bcab.2026.104207>
2. **Khan, S.A.**, Yadav, S.K. & Akhter, Y. (2026). A Comparative State-wise Assessment of Dengue Incidence and Deaths in India Using Annual Surveillance Data. *Discover Public Health* 23, article number 981. doi: 10.1186/s12982-026-02396-x

3. **Khan, S.A.,** Yadav, S.K. & Akhter, Y. (2026). Analyzing Global Threat of Lumpy Skin Disease Virus Infection: A Comparative Study of a Time Series Model and Machine Learning Models Across 10 High-Risk Countries. *Current Microbiology* 83, 389. doi: 10.1007/s00284-026-04982-3
4. Singh, V., Hajare, G. C., **Khan, S. A.,** & Kumari, A. (2026). Decoding socio-economic disparities in Uttar Pradesh: A spatio-temporal analysis using Wroclaw Taxonomy and K-means unsupervised machine learning clustering. *Mathematical Population Studies*, 1- 23. doi: 10.1080/08898480.2025.2612641
5. **Khan, S.A.,** Sharma, D.K., Shukla, A.K., Verma, M.R., Sharma, A., & Yadav, S.K. (2025). Traditional and Machine Learning Models for Forecasting Apple Production and Cultivation Area. *Applied Fruit Science* 67, 402. doi: 10.1007/s10341-025-01639-6
6. **Khan, S.A.,** Shukla, A.K., Yadav, S.K., & Vishwakarma, G.K. (2025). Machine learning models for analysis and prediction of optimal egg production. *Quality & Quantity*, 60, 1705. doi: 10.1007/s11135-025-02309-1.
7. Yadav, S.K., **Khan, S.A.,** Tiwari, M., Kumar, A., Kumar, V., & Akhter, Y. (2024). Taking cues from Machine Learning, Compartmental and Time Series models for SARS-CoV-2 Omicron infection in Indian Provinces. *Spatial and Spatio-Temporal Epidemiology*, 48, 100634. doi: 10.1016/j.sste.2024.100634.
8. Singh, V., **Khan, S.A.,** Yadav, S.K., & Akhter, Y. (2023). Modeling Global Monkeypox Infection Spread Data: A Comparative Study of Time Series Regression and Machine Learning Models. *Current Microbiology*, 81(1), Article 15. doi: 10.1007/s00284-023-03531-6.

National/International Conferences/Workshop

1. Research paper presented: **“Predicting Lumpy Skin Disease Outbreak: A Comparative Study of Machine Learning Across Affected Countries”** at **International Conference on Advances in Geometry algebra, Analysis and Artificial Intelligence (ICAGAAAI-2024)**”, organized by Department of Mathematics & Statistics, Integral University, Lucknow, India during November 7-9, 2024.
2. Research paper presented: **“A comparative study of Trends for Dengue outbreaks and mortality in India using different Machine Learning models”** at **International Conference on Innovations in Science, Management and Technology (ICISMT)**”, Organized By DEPARTMENT OF BUSINESS ADMINISTRATION Chaudhary Devi Lal University Sirsa, Haryana, India in Association with American Institute of Management and Technology (AIMT), USA (An Academic Subsidiary of The Global Knowledge foundation (GKF), USA, Inc.) during August 17-18, 2023.
3. Participated in the **“Workshop on Computational Oncology”**, organized by Department of Mathematics, Indian Institute of Technology Patna held during 19th April to 21st April, 2024.